

CODE ANALYSIS — SCHEMATIC (REV A) · P2-A-101 / A-102 REV F · P2-G-003 REV I

Design basis for the architect of record and the AHJ. Not a code review and no statement of approval: every line needs architect + AHJ confirmation (D-008). Values from KEYSTONE research notes; IBC 2021 text read on UpCodes (Alabama).

1 CODE BASIS + EDITIONS (R-006)

CODE	DESIGNED TO	COUNTY 2018 / STATE
Building	IBC 2021 (ASSUMED, Shane)	IBC 2018 (Madison County, since 2021-01-01)
Fire	IFC 2021 (State Fire Marshal; App. B, C, D)	IFC 2018 (county)
Life safety	not reviewed	NFPA 101 2018 (county); ch. 12 new assembly
Plumbing	IPC 2021 / IBC 2021 ch. 29 (ASSUMED)	IPC 2018; arena row of T2902.1 reads the same (R-009)
Assembly egress	IBC 2021 §1030.x	IBC 2018 §1029.x (same subsections; excerpts only, UNVERIFIED)
Energy / electrical	not checked	IECC 2015 (state) / NEC 2017

STATUS of every edition line: 2021 ASSUMED (Shane) — verify with AHJ (D-008 OPEN). Factors read in the 2021 text only; 2018 / NFPA 101 equivalence NOT verified (R-006.2).

AHJ: Madison County Inspection Department if the parcel is unincorporated (Hazel Green is; R-006.1, MEDIUM). DCM 'schoolhouse' review question OPEN. Assembly building needs a registered architect (Ala. Admin. Code 100-X-4-.03).

2 OCCUPANCY (R-007.1)

SPACE	GROUP	CITE (IBC 2021) / CONFIDENCE
Bowl + event floor	A-4	303.5 ('arenas'); HIGH
S&C, cross-training	A-3 or B	303.4 / 304.1 (training); MEDIUM
Admin, first aid	B	304.1
Storage, equipment, mech	S-1	R-007.1 (likely group)
Locker rooms, event lockers	A-4 or B	R-007.1
Mixed occupancy: nonseparated (508.3.1) with A-4 as the main occupancy, fully sprinklered — KEYSTONE recommendation, architect chooses (accessory 508.2 or separated 508.4 also possible).		

3 CONSTRUCTION TYPE

Construction type TBD (architect). Allowable height / area (IBC Tables 504.3, 504.4, 506.2) NOT checked. Inputs for that check: 2 stories, L2 FF 17'-9" (D-061), arena roof 42' ASSUMED, fully sprinklered. Drawn size (D-057, P2-A-101/102 Rev F): L1 53,080.6 / L2 28,932.6 / TOTAL 82,013.1 GSF (R-021 method).

4 FIRE PROTECTION + SYSTEMS (ASSUMED)

- Automatic sprinklers throughout (903.3.1.1, ASSUMED): 903.2.1.4 triggers for A-4 — fire area > 12,000 SF, occupant load >= 300, floor other than exit discharge; all three met (R-015.5).
- Emergency voice/alarm (907.2.1.1: Group A, occupant load >= 1,000; 907.5.2.2) ASSUMED (R-015.5).
- Sprinklers + voice alarm allow 0.2 in/occupant for stairs (1005.3.1 exc. 1) and 0.15 in for doors and other components (1005.3.2 exc.) (R-018.2, R-015.6).
- Smoke-protected seating NOT assumed (1030.6.2 needs smoke control per 909 + a life safety evaluation; NFPA 101 not reviewed). If used: 1030.6.2.2 roof >= 15' above the highest aisle (loop 17.75') = 32.75'; arena U/S structure 36' drawn (R-022 §5).
- Group A: post the occupant load (1004.9); fire safety + evacuation plan with a seating plan (IFC 403.2.1); telescopic seating per ICC 300 (1030.1.1) (R-007.2).

5 TRAVEL DISTANCE + COMMON PATH

- Exit access travel (T1017.2, A sprinklered): 250 ft. L2 worst measured ≈ 141 ft (R-018.4, Rev D plan; not re-measured on Rev F). L1: NOT CHECKED (TBD).
- Common path: seats 30 ft to two directions (1030.8; rows not drawn, TBD). Spaces with one exit: T1006.2.1 Group A 49 occupants / 75 ft — locker rooms (71 each) have 2 exits (X2 / X3 + corridor); event lockers (18 each) one door OK if the common path <= 75 ft (TBD).
- Stepped aisles, handrails, guards (1030.9, 1030.16, 1030.17): TBD with the seating layout (R-018.4-5).

6 OCCUPANT LOAD BY SPACE (IBC 2021 T1004.5 / 1004.6; R-007.2)

SPACE (areas drawn, P2-A-101/102 Rev F)	AREA SF	FACTOR	LOAD
LEVEL 1			
Lower tier seats (telescopic)		seats	1,100
Boys locker	3,507	50 gross	71
Girls locker	3,507	50 gross	71
Event lockers 1-4	3,600	50 gross	72
Flex / team assembly	2,008	15 net	134
Flex / storage (SW)	975	300 gross	4
Equipment storage + room	1,000	300 gross	4
Mech / elec (4 rooms)	3,881	300 gross	15
First aid	120	150 gross	1
Concession	275	200 gross	2

L1 without the event floor			1,474
Event floor — base case (sports)	16,416	50 gross	329

L1 TOTAL, base case (worst: table 7) 1,803

LEVEL 2			
Upper tier seats (fixed)		seats	1,100
Strength & conditioning	5,124	50 gross	103
Cross-training	3,500	50 gross	70
Admin	470	150 gross	4

L2 BASE			1,277
+ loop counted as exercise (R-018 a)	5,026	50 gross	101
+ standing at the W rail (R-018 b)	420	5 net	84

L2 WORST CASE (design, D-052) 1,462

Not added (same occupants moving through; R-018.1 method): lobby, concourses, athlete corridor / route, vestibule, restrooms, stairs. The AHJ may count waiting / standing space. Telescopic tier: number of seats (1004.6); if benches: 1 per 18 in of seat length (row lengths not drawn — may raise the count). Higher posted load by approval up to 1 per 7 SF with a seating diagram (1004.5.1). Event floor area 114 x 144 ft (R-007.2 / D-054; the locked program value is 16,400 SF, D-030).

7 EVENT FLOOR CASES (D-054 OPEN) — DESIGN TO THE WORST CASE

CASE (T1004.5)	FLOOR	L1 TOTAL	L2	BUILDING
Sports (exercise 50 gross, ASSUMED)	329	1,803	1,462	3,265
Tables + chairs (15 net)	1,095	2,569	1,462	4,031
Chairs only (7 net)	2,346	3,820	1,462	5,282
Standing (5 net) — WORST	3,284	4,758	1,462	6,220

L2 shown at its worst case (1,462, design basis, D-052) in every column. The building official assigns the floor load from the intended uses (1004.5); Shane decides the uses (D-054).

8 PLUMBING FIXTURE BASIS (R-009)

IPC 2021 ASSUMED (IBC 2021 T2902.1 assembly — arenas; 50/50 split 2902.1.1; urinals <= 67% of male WC, IPC 424.2). 2018 arena row reads the same (R-009). Fixture loads = P2-G-003 Rev I (seats + event floor at 50 gross).

LEVEL (FIXTURE LOAD)	WC M	WC F	LAV M	LAV F	DF	SS
L1 (G-003 Rev I: 1,428)	10	18	4	5	2	1
L2 (G-003 Rev I: 1,100)	8	14	3	4	2	1
L1 if the floor is standing (4,384)	26	50	11	15	5	1

G-003 Rev I counts the floor at 328 (16,400 SF ÷ 50). The standing case would need 16 more men's and 32 more women's WC on L1 than drawn — not drawn; follows D-054.

9 LEVEL 1 EXIT CAPACITY (doors 0.15 in/occ.)

Doors and other components 0.15 in per occupant (1005.3.2 exc.: sprinklers 903.3.1.1 + voice alarm 907.5.2.2). Widths are NOT drawn on P2-A-101 Rev F: each door symbol taken as a pair of 3'-0" leaves (A-201: 6' x 8' symbols, ASSUMED) at 2 x 32 in = 64 in clear (1010.1.1: 32 in minimum clear per leaf) — architect sets the real widths. Provided: E1 + X1-X10 = 11 openings x 64 in = 704 in. Stair discharge doors X4 / X6 / X9 / X10 carry the L2 worst case first (1,462 ÷ 4 x 0.15 = 54.8 in each).

CHECK (in)	329	1,095	2,346	3,284	PROV.
Total at grade (L1 + L2)	489.8	604.6	792.3	933.0	704.0
Main exit E1 >= 1/2 (1030.2)	244.9	302.3	396.1	466.5	64.0
Other L1 exits >= 1/2 L1 (1030.3)	135.2	192.7	286.5	356.8	420.7
Lose one opening >= 50% (1005.5)	244.9	302.3	396.1	466.5	640.0

Columns = event-floor load (D-054 cases). Green = enough as calculated; red = SHORT. 4 exits needed per story over 1,000 (T1006.3.3): 11 on L1.

10 FINDING — L1 DOORS vs THE WORST CASE

SHORT at the worst case (standing floor, 3,284; building 6,220): main exit E1 needs 466.5 in = 8 pairs vs 1 pair (64 in): −402.5 in. Total door capacity needs 933.0 in vs 704 in: −229.0 in (≈ 4 pairs). Other L1 exits pass (1030.3). Even the sports case needs E1 >= 244.9 in (4 pairs).

- OPTION 1 — Main-exit door bank at E1 (keeps the single controlled entry, D-033): 8 pairs = 512 in across the lobby's south wall (58 ft of frontage; first aid / vestibule edges move — architect). Total then 1,152 in >= 933.0 in. Security lanes stay out of the egress width (1003.6, 1010.5).
- OPTION 2 — Multiple main exits around the perimeter (1030.2 / 1030.3 last sentences; total >= 100%): 15 pairs total = 960 in, i.e. add 4 pairs (e.g. widen E1 and X5 / X7 / X8 to double pairs). Needs the AHJ to accept that E1 is not the one well-defined main exit; more exterior doors to alarm and staff.

No sheet redrawn: door widths are set by the architect once D-054 is answered.

11 LEVEL 2 EGRESS (R-018, D-052)

Stairs 0.2 in per occupant (1005.3.1 exc. 1): 4 stairs x 76 in = 304 in. Base 1,277 x 0.2 = 255.4 in; worst 1,462 x 0.2 = 292.4 in → spare 11.6 in (passes as calculated). Lose one stair: 228 in >= 146.2 in (1005.5). 4 exits (T1006.3.3). Loop 84 in vs 78.6 in at the worst corner (R-018.3 method, Rev F seat split). Intermediate handrails on 76 in stairs (1014.9; architect to confirm).

12 OPEN ITEMS FOR THE ARCHITECT / AHJ

- Code editions + AHJ (D-008): 2021 vs 2018 / NFPA 101 — pre-application meeting with Madison County.
- Event floor uses (D-054): sets the L1 exits, main exit and L1 fixtures (tables 7-10).
- Door widths, hardware, main-exit frontage (1030.2: street or 10 ft open space) — architect.
- Construction type + allowable area (ch. 5 / 6) — architect.
- Smoke-protected seating or not (1030.6.2) — architect / AHJ.
- Upper-tier structure 1'-6" ASSUMED (structural engineer): front-row clear is exactly 7'-6" (1003.2), zero margin.
- Seat type on the telescopic tier (benches: 1 per 18 in) and the seating layout (aisles, rails, guards).

PRELIMINARY — CONCEPTUAL DESIGN — NOT FOR CONSTRUCTION

PROJECT	PHASE	SHEET TITLE	SCALE	DATE	REV	DRAWN BY	SHEET
Hazel Green Regional Athletic Complex Trojan Horse Arena · Hazel Green area, Madison County, Alabama (site TBD, D-006)	PHASE 2	CODE ANALYSIS SCHEMATIC	NTS	October 2026	A	Drawn by KEYSTONE (AI) for Shane Brazelton	P2-G-002